

Supplementary Information

Explainable AI and pharmacogenomics for opioid and polypharmacy risk

R. Jerome Dixon and Elvin T. Price

This file is the single Supplementary Information PDF for the Article (systematic review submitted as Article per *npj Genomic Medicine* content types).

File	Contents
S1	Nine PubMed Boolean search strings by OODA-phase keyword domain
S2	PRISMA 2020 checklist with manuscript section cross-references
S3	Data extraction for OODA Decide-phase studies with NIH AI Reporting Checklist ratings (CSV; 142 studies). Deposited at https://github.com/Jerome3590/pgx-analysis because a 20-column extraction table is not practical as paginated PDF.
S4	PROBAST risk-of-bias assessments for topic-relevant prediction model studies ($n = 73$)
S5	Full evidence map (6,182 included articles with OODA-phase and CRISP-DM classification). Deposited at https://github.com/Jerome3590/pgx-analysis because of file size.
S6	Outcome-stratum labels for the 94-study ML prediction subset (below)

No patient-level claims data are included.

File S6. Outcome strata in the ML prediction subset ($n = 94$)

Primary outcome category	n	% of 94	Notes
OUD diagnosis / dependence	43	46%	Modal ICD-10 F11.20
Opioid-related ED or overdose utilization	27	29%	Claims/EHR utilization definitions
Overdose (ICD-10 T40.xx) as primary label	19	20%	Overdose as the primary outcome

Primary outcome category	<i>n</i>	% of 94	Notes
Polypharmacy ADE or DDI	5	5%	ADE/DDI-related primary outcome

Supplementary File S1

Full PubMed Boolean search strings for the Systematic Quantitative Literature Review (SQLR). Searches were conducted via PubMed API (January 2015–March 2026) across nine targeted strings organised by OODA-phase keyword ontology.

String 1 — Opioid ED Risk (core)

("opioid use disorder"[MeSH] OR "opioid"[tiab] OR "OUD"[tiab]) AND ("emergency department"[tiab] OR "ED visit"[tiab] OR "emergency room"[tiab]) AND ("machine learning"[tiab] OR "predictive model"[tiab] OR "risk prediction"[tiab])

String 2 — Polypharmacy & ADE

("polypharmacy"[tiab] OR "adverse drug event"[MeSH] OR "drug-drug interaction"[tiab]) AND ("machine learning"[tiab] OR "artificial intelligence"[tiab] OR "risk score"[tiab]) AND ("claims"[tiab] OR "EHR"[tiab] OR "electronic health record"[tiab])

String 3 — APCD / Claims Data

("all-payer claims"[tiab] OR "all payer claims database"[tiab] OR "APCD"[tiab] OR "Medicare"[MeSH] OR "Medicaid"[MeSH]) AND ("opioid"[tiab] OR "polypharmacy"[tiab]) AND ("prediction"[tiab] OR "modeling"[tiab])

String 4 — Pharmacogenomics (PGx)

("pharmacogenomics"[MeSH] OR "pharmacogenetics"[tiab] OR "PGx"[tiab] OR "CYP2D6"[tiab] OR "OPRM1"[tiab] OR "gene-drug interaction"[tiab]) AND ("opioid"[tiab] OR "polypharmacy"[tiab] OR "adverse drug"[tiab])

String 5 — CPIC & Gene-Drug Guidelines

("CPIC"[tiab] OR "Clinical Pharmacogenomics Implementation Consortium"[tiab] OR "metabolizer phenotype"[tiab] OR "poor metabolizer"[tiab]) AND ("machine learning"[tiab] OR "clinical decision support"[tiab] OR "risk"[tiab])

String 6 — XAI / Explainability

("explainable artificial intelligence"[tiab] OR "explainable AI"[tiab] OR "SHAP"[tiab] OR "LIME"[tiab] OR "feature importance"[tiab] OR "interpretable machine learning"[tiab]) AND ("opioid"[tiab] OR "polypharmacy"[tiab] OR "pharmacogenomic"[tiab] OR "clinical"[tiab])

String 7 — Gradient Boosting Methods

("gradient boosting"[tiab] OR "XGBoost"[tiab] OR "CatBoost"[tiab] OR "random forest"[tiab] OR "LightGBM"[tiab]) AND ("opioid"[tiab] OR "polypharmacy"[tiab] OR "adverse drug event"[tiab] OR "pharmacogenomic"[tiab])

String 8 — Causal / Temporal Modeling

("causal inference"[tiab] OR "target leakage"[tiab] OR "temporal validation"[tiab] OR "dynamic time warping"[tiab] OR "DTW"[tiab] OR "process mining"[tiab]) AND ("opioid"[tiab] OR "polypharmacy"[tiab] OR "clinical prediction"[tiab])

String 9 — Clinical Decision Support Deployment

("clinical decision support"[MeSH] OR "CDS"[tiab] OR "point-of-care"[tiab] OR "serverless"[tiab] OR "cloud"[tiab]) AND ("pharmacogenomic"[tiab] OR "opioid risk"[tiab] OR "polypharmacy risk"[tiab]) AND ("machine learning"[tiab] OR "artificial intelligence"[tiab])

Section	Item #	Checklist Item	Location in Manuscript
TITLE	1	Identify the report as a systematic review.	Title
ABSTRACT	2	See the PRISMA 2020 for Abstracts checklist.	Abstract
INTRODUCTION	3	Describe the rationale for the review.	§1.1
INTRODUCTION	4	Provide an explicit statement of the objectives.	§1.4
METHODS	5	Specify the inclusion and exclusion criteria.	§2.2
METHODS	6	Specify all databases, registers, websites, and other sources searched.	§2.3
METHODS	7	Present the full search strategies for all databases.	File S1
METHODS	8	Specify the methods used to decide whether a study met the inclusion criteria.	§2.2
METHODS	9	Specify the methods used to collect data from reports.	§2.4
METHODS	10	List and define all outcomes for which data were sought.	§2.2

Section	Item #	Checklist Item	Location in Manuscript
METHODS	11	Describe the methods used to assess risk of bias.	§2.5
METHODS	12	Specify all measures used to synthesize results.	§2.6
METHODS	13	Describe any methods to identify selective reporting.	§2.5
METHODS	14	Describe any sensitivity analyses conducted.	§2.6
METHODS	15	Describe any methods to assess certainty of evidence.	§2.5
RESULTS	16	Describe the results of the search and selection process (PRISMA flow diagram).	§3.1 / Fig. 1
RESULTS	17	Cite each included study and present its characteristics.	§3.2 / File S3
RESULTS	18	Present assessments of risk of bias for each included study.	§3.3 / File S4
RESULTS	19	For all outcomes, present a summary of available data.	§3.2–3.4
RESULTS	20	Present results of all statistical syntheses.	§3.4

Section	Item #	Checklist Item	Location in Manuscript
RESULTS	21	Present results of any investigations of causes of heterogeneity.	§3.4
DISCUSSION	22	Provide a general interpretation of the results.	§4
DISCUSSION	23	Discuss any limitations of the evidence.	§4.3
DISCUSSION	24	Describe the implications of the results.	§4.4
OTHER	25	Provide registration information for the review.	Abstract (PROSPERO)
OTHER	26	Indicate where the review protocol can be accessed.	Not pre-registered
OTHER	27	Describe sources of financial or non-financial support.	Funding

Table S2. PRISMA 2020 Checklist.

PROBAST									
Selec-tion / Rele-vance									
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
2936	Davis (2025)	Utility of Can- di- date Genes From an Algo- rithm De- signed to Pre- dict Ge- netic Risk for Opi- oid Use Dis- orde	L	U	L	H	H	Low	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
5770	Yu (2023)	DGANDI: Dou- ble Gen- era- tive Ad- ver- sarial Net- works for Drug- Drug Inter- ac- tion Pre- dic- tion.	U	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Drug- Drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Selec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
5142	Dafniet (2024)	Prediction of ad- verse drug reac- tions due to ge- netic pre- dis- posi- tion using deep neu- ral net- works.	U	H	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'ad- verse drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

PROBAST									
Selec- tion / Rel- e- vance									
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
8483	Parton (2026)	A ma- chine learn- ing ap- proach for opi- oid over- dose risk pre- dic- tion among Al- abama Medi- caid bene- ficia- ries	L	L	L	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

PROBAST								
Selec- tion / Rel- e- vance								
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern
Path- way								
5129	Hu (2024)	MPHGCL- DDI: Meta- Path- Based Het- ero- ge- neous Graph Con- trastive Learn- ing for Drug- Drug Inter- ac- tion Predi	L	L	H	L	H	Low
Included: CRISP-DM modeling + NIH AI tags; ti- tle/notes match 'Drug-Drug' (OUD/opioid-ED/polypharmacy-ADE/PGx/medication-ADE rele- vance). OODA phase=act; CRISP-DM=modeling.								

								PROBAST	
								Se- lec- tion / Rel- e-	
	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Overall Con- cern	Path- way
5080	Wang (2025)	BiRNN-U DDI: A Drug- Drug Inter- ac- tion Event Type Pre- dic- tion Model Based on Bidi- rec- tional Re- cur- rent Neur		U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Drug- Drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST	
Study_	Year	First Au- thor (trun- cated)	Title	D1	D2	D3	D4	Overall ROB	Overall Con- cern	Se- lec- tion / Rel- e- vance Path- way
4921	Gao (2024)	Precision Ad- verse Drug Reac- tions Pre- dic- tion with Het- ero- ge- neous Graph Neu- ral Net- work.	L	U	L	H	H	Low	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Ad- verse Drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.	

									PROBAST
									Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
6012	Noor (2022)	A Data- Driven Medi- cal Deci- sion Frame- work for Asso- ciat- ing Ad- verse Drug Events with Drug- Drug Inter- act	U	U	H	L	H	Low	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Ad- verse Drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

								PROBAST
								Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern
5263	Gao (2024)	AutoDDU Drug- Drug Inter- ac- tion Pre- dic- tion With Auto- mated Graph Neu- ral Net- work.	U	U	H	H	H	Moderate
								Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Drug- Drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

PROBAST								
Selec- tion / Rel- e- vance								
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern Path- way
6155	Moore (2021)	MachineU Learn- ing to Iden- tify Inter- ac- tion of Single- Nucleotide Poly- mor- phisms as a Risk Fac- tor for Chr	L	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Drug- Induced' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Selec- tion / Rel- e-
									vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
5926	Yu (2022)	STNN- DDI: a Substructure- aware Ten- sor Neu- ral Net- work to pre- dict Drug- Drug Inter- ac- tions.	U	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Drug- Drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

								PROBAST
								Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern
								Path- way
3463	Giladi (2023)	Patient-U Reported Data Aug- ment Pre- dic- tion Mod- els of Per- sis- tent Opi- oid Use after Elec- tive Up- per Ex- trem		U	H	H	H	Moderate Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
2169	Kashyap (2023)	A deep learn- ing method to de- tect opi- oid pre- scrip- tion and opi- oid use disor- der from elec- tronic health	L	U	L	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
8493	Xue (2025)	Development and eval- ua- tion of a ma- chine learn- ing model to pre- dict acute care for opi- oid use disor- der	Ident	U	L	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

								PROBAST	
								Se- lec- tion / Rel- e-	
Study_	First	Title	D1	D2	D3	D4	Overall	Path-	
	Au- thor (Year)	(trun- cated)					Concern		way
5181	Geng (2024)	MGDDIU A multi- scale graph neu- ral net- works for drug- drug inter- ac- tion pre- dic- tion.	U		H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'drug- drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
3370	Shah- Mohammadi (2023)	Identification of Sub- phe- no- types of Opi- oid Use Dis- order Us- ing Un- su- per- vised Ma- chine Learn- ing.	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.	

									PROBAST
									Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
5098	Kang (2024)	A Dual- Modality Complex- Valued Fu- sion Method for Pre- dict- ing Side Ef- fects of Drug- Drug Inter- ac- tions B	U	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Drug- Drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

PROBAST									
Selec- tion / Rel- e- vance									
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
4700	Sicard (2025)	Can large lan- guage mod- els de- tect drug- drug inter- ac- tions lead- ing to ad- verse drug reac- tions?	U	U	L	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'drug- drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Selec- tion / Rel- e-
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Overall Con- cern	vance Path- way
5960	He (2022)	3DGT- DDI: 3D graph and text based neu- ral net- work for drug- drug inter- ac- tion pre- dic- tion.	U	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'drug- drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

PROBAST									
Selec- tion / Rel- e- vance									
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
6523	Shankar (2021)	Predicting ad- verse drug reac- tions of two- drug com- bina- tions using struc- tural and tran- scrip- tomic drug	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'ad- verse drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.	

										PROBAST
										Se- lec- tion / Rel- e-
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Overall Con- cern	Vance Path- way	
4524	Wong (2025)	DDI- CYP: Metabolism en- sem- ble mod- els for drug- drug inter- ac- tion pre- dic- tions.	U	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'drug- drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.	

PROBAST								
Selec- tion / Rel- e- vance								
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Overall Con- cern
Path- way								
2389	Lo- Ciganic hu- (2021)	Integrating man ser- vices and crim- inal jus- tice data with claims data to pre- dict risk of opi- oid over	High	L	L	L	U	Moderate
	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.							

								PROBAST	
								Selec- tion / Rel- e-	
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Overall Con- cern	Path- way
3492	Castle (2023)	MachineU Learn- ing Model Iden- tifies Pre- oper- ative Opi- oid Use, Male Sex, and Ele- vated Body Mass In- dex as	U	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

								PROBAST
								Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern Path- way
3764	Klemt (2022)	MachineU learn- ing algo- rithms pre- dict ex- tended post- oper- ative opi- oid use in pri- mary total knee arthropl	U	U	H	H	H	Moderate Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

								PROBAST
								Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern
								Path- way
3386	Lu (2023)	Editorial Com- men- tary: Ma- chine Learn- ing and Arti- ficial Intel- li- gence Are Tools Re- quir- ing Physi- cian and	U	U	H	H	H	Moderate Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

								PROBAST
								Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern
								Path- way
9141	Dong (2021)	Identifying risk of opi- oid use disor- der for pa- tients tak- ing opi- oid medi- ca- tions with deep learn- ing.	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Se- lec- tion / Rel- e-
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Overall Con- cern	Path- way
4494	Wang (2025)	A novel mem- ory inter- ac- tion neu- ral net- work for multi- label drug- drug inter- ac- tion pre- dic- tion with neig	U	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'drug- drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Se- lec- tion / Rel- e-
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Overall Con- cern	Path- way
4888	Xu (2024)	DDI- GPT: Ex- plain- able Pre- dic- tion of Drug- Drug Inter- ac- tions using Large Lan- guage Mod- els en- hanced with	U	U	H	L	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Drug- Drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Selec- tion / Rel- e-
									vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
5341	Zhao (2023)	Drug- drug inter- ac- tion pre- dic- tion: databases, web servers and com- puta- tional mod- els.	L	U	H	L	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Drug- drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Selec- tion / Rel- e-
									vance
									Path-
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	way
6207	Schwarz (2021)	Attention- Siamese attention- based deep learn- ing method for drug- drug inter- ac- tion pre- dic- tions.	U	U	L	L	U	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'drug- drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

PROBAST									
Selec- tion / Rel- e- vance									
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
3702	Bemel (2022)	A Pre- dic- tive Model for De- vel- op- ing Long Term Opi- oid Use After Neu- ro- surgery and Or- tho- pedic Surgery.	U	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

PROBAST								
Selec- tion / Rel- e- vance								
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern
Path- way								
5952	Feng (2022)	Prediction of Drug- Drug Inter- ac- tion Us- ing an Attention- Based Graph Neu- ral Net- work on Drug Molec- ular	H	U	H	L	H	Low
Included: CRISP-DM modeling + NIH AI tags; ti- tle/notes match 'Drug-Drug' (OUD/opioid-ED/polypharmacy-ADE/PGx/medication-ADE rele- vance). OODA phase=act; CRISP-DM=modeling.								

									PROBAST
									Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
4935	Jeong (2025)	Discovering Se- vere Ad- verse Reac- tions From Phar- ma- coki- netic Drug- Drug Inter- ac- tions Through Liter- ature	U	L	H	H	Low	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Drug- Drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.	

									PROBAST
									Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
5741	Lin (2023)	R2- DDI: relation- aware fea- ture re- fine- ment for drug- drug inter- ac- tion pre- dic- tion.	U	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'drug- drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

PROBAST									
Selec- tion / Rel- e- vance									
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
8989	Dong (2023)	An inte- grated LSTM- HeteroRGNN model for inter- pretable opi- oid over- dose risk pre- dic- tion.	U	U	H	L	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
4502	Shi (2025)	A Dose- Aware Model for Re- veal- ing Dose- Risk Rela- tion- ship of Drug- Drug Inter- ac- tion.	L	L	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Drug- Drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

PROBAST									
Selec- tion / Rel- e- vance									
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
5922	He (2022)	Multi- type fea- ture fu- sion based on graph neu- ral net- work for drug- drug inter- ac- tion pre- dic- tion.	U	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'drug- drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Selec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
4626	Bermúdez Pérez (2025)	Evaluation of the role of comed- ica- tion prop- erties in the sever- ity of drug- induced liver in- jury using		U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'drug- induced' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

PROBAST								
Selec- tion / Rel- e- vance								
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Overall Con- cern
Path- way								
5778	Gill (2023)	Evaluating the per- for- mance of machine- learning re- gres- sion mod- els for phar- ma- coki- netic drug- drug inter	U	L	L	U	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'drug- drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

								PROBAST
Study_	First	Title	D1	D2	D3	D4	Overall	Selec- tion / Rel- e- vance
	Author (Year)	(trun- cated)					ROB	Path- way
4702	Hammar (2025)	MachineU Learn- ing to Im- prove Deci- sion Sup- port for Pre- vent- ing Ad- verse Drug Events.	U	U	H	H	H	Moderate Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Ad- verse Drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

PROBAST								
Selec- tion / Rel- e- vance								
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern Path- way
3021	Simpson (2024)	Classifying High- Risk Pa- tients for Per- sis- tent Opi- oid Use After Ma- jor Spine Surgery: A Machine- Learni		U	H	H	H	Moderate Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

PROBAST									
Selec- tion / Rel- e- vance									
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
6287	Luo (2021)	Novel deep learning- based tran- scrip- tome data anal- ysis for drug- drug inter- ac- tion pre- dic- tion with an a	U	U	L	H	H	Low	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'drug- drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

								PROBAST	
								Selec- tion / Rel- e- vance	
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
5620	Nguyen (2024)	Central-U Smoothing Hy- per- graph Neu- ral Net- works for Pre- dict- ing Drug- Drug Inter- ac- tions.		U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Drug- Drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
5159	Abdul Ra- heem (2024)	A Novel Deep Learn- ing Model for Drug- drug Inter- ac- tions.	U	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Drug- drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
1817	Tahir (2025)	Enhanced drug- drug inter- ac- tion ex- trac- tion from biomed- ical text using deep learning- based sen- tence re	U	U	L	L	U	Low	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'drug- drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
786	Farooq (2026)	Explaining machine learning for predicting opioid-related aberrant behavior: A multi-modal approach	U	H	L	H	Moderate	Included: CRISP-DM modeling + NIH AI tags; title/notes match 'opioid' (OUD/opioid-ED/polypharmacy-ADE/PGx/medication-ADE relevance). OODA phase=decide; CRISP-DM=modeling.	

									PROBAST
									Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
3502	Warren (2022)	Using ma- chine learn- ing to study the effect of medi- ca- tion ad- her- ence in Opi- oid Use Dis- or- der.	U	L	L	H	H	Low	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
5925	Jeong (2022)	Detecting Drug- Drug Inter- ac- tions in COVID- 19 Pa- tients.	U	H	H	H	Low		Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Drug- Drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
5901	Nguyen (2022)	SPARSE: a sparse hy- per- graph neu- ral net- work for learn- ing mul- tiple types of la- tent com- bina- tions to acc	H	U	H	L	H	Low	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'drug- drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

PROBAST								
Selec- tion / Rel- e- vance								
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern Path- way
5550	Akimoto (2023)	Detection of po- ten- tial drug- drug inter- ac- tions for risk of acute kid- ney in- jury: a population- based ca	h	L	L	L	L	Moderate Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'drug- drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

PROBAST								
Selec- tion / Rel- e- vance								
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Overall Con- cern
Path- way								
5305	Kpanou (2024)	Learning self- supervised molec- ular rep- re- sen- ta- tions for drug- drug inter- ac- tion pre- dic- tion.	U	U	L	L	U	Low
Included: CRISP-DM modeling + NIH AI tags; ti- tle/notes match 'drug-drug' (OUD/opioid-ED/polypharmacy-ADE/PGx/medication-ADE relevance). OODA phase=act; CRISP-DM=modeling.								

									PROBAST
									Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
2991	Robinson (2024)	Spatial L pat- terns of rural opioid- related hos- pital emer- gency de- part- ment vis- its: A ma- chine learn- ing an		U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

PROBAST								
Selec- tion / Rel- e- vance								
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern
Path- way								
2814	Jones (2025)	Identifying key risk fac- tors for in- ten- tional self- harm, in- clud- ing sui- cide, among a co- hort of peo- ple	U	L	L	U	Low	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

										PROBAST
										Se- lec- tion / Rel- e-
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Overall Con- cern	Overall Path- way	
6158	Lin (2022)	MDF- SA- DDI: pre- dict- ing drug- drug inter- ac- tion events based on multi- source drug fu- sion, multi- source	U	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'drug- drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.	

PROBAST									
Selec- tion / Rel- e- vance									
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
4403	Jia (2025)	A meta-contrastive learning approach for clinical drug-drug interaction extraction from biomedical l	U	L	L	L	U	Moderate	Included: CRISP-DM modeling + NIH AI tags; title/notes match 'drug-drug' (OUD/opioid-ED/polypharmacy-ADE/PGx/medication-ADE relevance). OODA phase=act; CRISP-DM=modeling.

									PROBAST
									Selec- tion / Rel- e-
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Overall Con- cern	vance Path- way
6151	Nyamab (2022)	Drug- drug inter- ac- tion pre- dic- tion with learn- able size- adaptive molec- ular sub- struc- tures.	U	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Drug- drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
4294	Chen (2026)	Knowledge- graph- enhanced multi- scale mod- eling for drug- drug inter- ac- tion pre- dic- tion.	ge-	U	L	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'drug- drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

PROBAST								
Selec- tion / Rel- e- vance								
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Overall Con- cern
Path- way								
3673	Dunham (2022)	Evaluation of an Opi- oid Over- dose Com- pos- ite Risk Score Cut- off in Ac- tive Duty Mili- tary Ser- vice Mem- bers	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

PROBAST								
Selec- tion / Rel- e- vance								
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Overall Con- cern
Path- way								
4721	Ren (2025)	Predicting rare drug- drug inter- ac- tion events with dual- granular structure- adaptive and pair varia- tio	U	H	L	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'drug- drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
4492	Mondal A (2025)	sys- tem- atic map- ping re- view on the capa- bility of large lan- guage mod- els in drug- drug inter- ac- tion anal	U	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'drug- drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Selec- tion / Rel- e-
									vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
2018	Chen (2024)	Automatic text clas- sifica- tion of drug- induced liver in- jury using document- term ma- trix and XG- Boost.	U	U	L	L	U	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'drug- induced' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=decide; CRISP- DM=modeling.

									PROBAST
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Selec- tion / Rel- e- vance
									Path- way
1777	Prioleau (2026)	Leveraging Large Lan- guage Mod- els for Ad- verse Drug Event De- tec- tion: A Com- para- tive Study of To- ken and	L	H	H	H	Low	Included: CRISP-DM modeling + NIH AI tags; ti- tle/notes match 'Ad-verse Drug' (OUD/opioid-ED/polypharmacy-ADE/PGx/medication-ADE relevance). OODA phase=act; CRISP-DM=modeling.	

PROBAST								
Selec- tion / Rel- e- vance								
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern
4834	Luo (2025)	Exploiting question- answer frame- work with multi- GRU to de- tect ad- verse drug reac- tion on so- cial me- dia.		L	L	H	H	Low
Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'ad- verse drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.								

									PROBAST
									Selec- tion / Rel- e-
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Overall Con- cern	vance Path- way
6118	El- Allaly (2022)	An at- ten- tive joint model with transformer- based weighted graph con- volu- tional net- work for ex- tract- ing	U	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'ad- verse drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

								PROBAST
								Selec- tion / Rel- e-
Study_	First	Title	D1	D2	D3	D4	Overall	vance
	Au- thor (Year)	(trun- cated)					ROB	Path- way
4989	Hou (2024)	MSDAFL: molec- ular substructure- based dual at- ten- tion fea- ture learn- ing frame- work for pre- dict- ing drug- d		U	L	L	U	Moderate
								Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'drug- drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

										PROBAST
										Selec- tion / Rel- e-
										vance
										Path-
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	way	
2379	Kural (2024)	Using ma- chine learn- ing to im- prove ana- phy- laxis case iden- tifica- tion in medi- cal claims data.	U	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'ana- phy- laxis' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=decide; CRISP- DM=modeling.	

PROBAST								
Selec- tion / Rel- e- vance								
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Overall Con- cern
Path- way								
8492	Sharp (2025)	Identifying emer- gency de- part- ment pa- tients at high risk for opi- oid over- dose using natu- ral lan- guage pr	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
2793	Ahmad (2025)	Sentiment Anal- ysis Us- ing a Large Lan- guage Model- Based Ap- proach to De- tect Opi- oids Mixed With Other Su	U	U	L	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

								PROBAST
Study_	First	Title	D1	D2	D3	D4	Overall	Selec- tion / Rel- e- vance
	Au- thor (Year)	(trun- cated)					ROB	Path- way
3637	Stafford (2022)	Identifying key risk fac- tors for pre- ma- ture dis- con- tinu- ation of opi- oid use disor- der treat- ment in the U	U		H	H	H	Moderate Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Selec- tion / Rel- e-
									vance
									Path-
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	way
804	Lo (2026)	Develop- ment and vali- da- tion of ma- chine learn- ing mod- els to pre- dict vancomycin- and teicoplanin- associa	Ident	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'van- comycin' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=decide; CRISP- DM=modeling.

									PROBAST
									Se- lec- tion / Rel- e- vance
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	Path- way
6396	Gonzalez Jimenez (2021)	Drug prop- erties and host fac- tors con- tribute to bio-chem- ical pre- sen- ta- tion of drug- induced liver in- jury	U	U	H	H	H	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'drug- induced' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

									PROBAST
									Selec- tion / Rel- e-
									vance
									Path-
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern	way
2644	Pagare (2025)	Augmen- ling large lan- guage mod- els to pre- dict so- cial de- ter- mi- nants of men- tal health in opi- oid use disor	U	L	L	U	Moderate	Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'opi- oid' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.	

PROBAST								
Selec- tion / Rel- e-								
vance								
Path-								
way								
Study_	First Au- thor (Year)	Title (trun- cated)	D1	D2	D3	D4	Overall ROB	Con- cern
4798	Zaikis (2025)	TransformDDI: U The Transformer- Based Joint Multi- Task Model for End- to- End Drug- Drug Inter- ac- tion Extr						
								Included: CRISP- DM mod- eling + NIH AI tags; ti- tle/notes match 'Drug- Drug' (OUD/opioid- ED/polypharmacy- ADE/PGx/medication- ADE rele- vance). OODA phase=act; CRISP- DM=modeling.

Table S4. PROBAST risk of bias and applicability assessments for 73 topic-relevant prediction-model studies. Selection filter: CRISP-DM phase = modeling, NIH AI checklist score present, and title/notes match OUD, opioid-related ED/overdose, polypharmacy ADE/DDI, PGx, or medication-related ADE terms. Off-topic Decide+modeling articles (e.g., unrelated imaging or sensor applications) are excluded even when NIH AI tags are present. OODA Decide is not required because many topic-relevant prediction papers are mapped to Act:modeling in the evidence map. D1 Participants, D2 Predictors, D3 Outcome, D4 Analysis. L = Low, H = High, U = Unclear. Domain ratings derived from NIH AI check-

list tags. Each row includes a study-specific PROBAST Selection / Relevance Pathway string.